

M. H. Pesaran (2015): Time series and panel data econometrics. Oxford University Press, Oxford, 1104 pp, Hardcover 110.00 £, ISBN: 9780198736912

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This is an extensive exposition of modern time series econometrics and panel data analysis reflecting in particular Hashem Pesaran's many contributions in these areas over the last decades. The book has four distinct features. First, it is rather self-contained, beginning with an introductory part on econometrics, providing statistical prerequisites, covering classical univariate as well as multivariate time series analysis and different aspects of panel data econometrics; further, it is complemented by appendices on mathematics and probability. Second, the book includes recent developments on e.g. forecasting (like forecast evaluation criteria or density forecasts), on model selection (Bayesian approach), or on spatial panel econometrics, which are not covered by competing textbooks on time series or panel data. Third, the book contains, in a unified framework, topics that are closely associated with Hashem Pesaran, e.g. generalized impulse response functions, panel cointegration, or global VARs. Fourth, this is a textbook with a clear focus on econometrics, dedicating a lot of attention to e.g. cointegration and with special chapters on the efficient market hypothesis or rational expectations models (including DSGE); moreover, all empirical examples are from economics and finance. The book has too many chapters (33 plus 3 appendices) to review them in detail here. It begins with Part I on econometrics covering 160 pages. Part II deals with statistical theory on 100 pages and treats e.g. generalized method of moments in one chapter. These two parts make the book rather self-contained for students of econometrics. Part III has two short chapters on stochastic processes, one of them being settled in the frequency domain; it does not treat the multivariate case. Part IV is dedicated to univariate time series analysis on 135 pages. The multivariate Part V stretches over 200 pages. The reviewer appreciates in particular Chap. 19, which is

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a systematic introduction to multivariate techniques from seemingly unrelated regression estimation over factor models to canonical correlation analysis. The final Part VI is almost a textbook on its own. It is the largest part on panel econometrics with 300 pages with chapters on classical panel models (short T), on large panels ($T \rightarrow \infty$), on cross-sectional dependence, and on nonstationary panels.

Given the length of the book it is not surprising that it is not totally homogeneous in terms of the level of rigor. Some chapters contain proofs of some elementary results (Chap. 8 on asymptotic theory, or Appendix B on probability), while other chapters only present results established in other books (e.g. Chap. 15 on univariate unit roots). Hence, parts of the book may be used for undergraduate teaching, while others are more suitable for PhD students. At the end of each chapter there is a short section with exercises—without solutions. Given the broad scope of the exposition with so many up-to-date research fields synthesized in a unified framework, the book will serve in particular as a valuable reference manual for scholars engaged in applied econometrics.